

FUNCTIONAL REQUIREMENTS DOCUMENT

LODESTAR

Autonomous Logistics Exception Agent

Understand the mess. Act on it. Prove it moved.

Version 1.0 · Status: Built & Verified · Citta RISE — Idea2Agent, Problem 06

1. Purpose

This document defines what LODESTAR does, function by function. LODESTAR is an AI agent that takes a raw logistics-exception input, resolves it by performing real actions on a logistics system, and proves the outcome by re-reading system state — retrying transient failures, escalating permanent ones, and never claiming success it cannot prove.

2. Scope

- In scope: ingest a raw exception artifact, extract the shipment, diagnose, dynamically choose tools, act, verify, and notify or escalate — plus an operator console (run, browse shipments, history, insights).
- Out of scope: production carrier/TMS integrations, user accounts/auth, a persistent database, and multi-tenancy. v1 uses a simulated logistics system plus one real geocoding call.

3. Actors

Actor	Role
Operator	Runs the agent, reviews results, approves high-value actions, browses shipments/history.
AI Agent	Autonomously plans and executes the resolution.
External systems	LLM (Gemini → Groq), OpenStreetMap Nominatim (geocoding), Resend / Twilio (notifications).

4. Functional Requirements

4.1 Input & Understanding

FR-1 The system shall accept a raw exception artifact as text (email), JSON (webhook), or an image (photo of a label).

FR-2 The system shall extract the shipment ID and exception type from the artifact, handling messy or unstructured input.

FR-3 For image input, the system shall use vision OCR to read the text before extraction.

FR-4 The system shall use a deterministic extractor first and invoke the LLM only when fixed rules are uncertain.

4.2 Context Gathering

FR-5 The system shall look up authoritative shipment state (status, carrier, hub, ETA, exception) plus the linked order and customer.

FR-6 The system shall retrieve supporting context on demand (available carriers by region, inventory).

4.3 Dynamic Resolution

FR-7 The agent shall select tools based on the situation, not a fixed sequence; the full tool set is available on every turn.

FR-8 The system shall perform real state-changing actions: reroute, reschedule delivery, update address, reassign carrier, issue credit, and escalate.

FR-9 The system shall validate a delivery address against live external geocoding before updating it.

FR-10 State-changing actions shall be idempotent — a repeat of the same action does not double-apply.

4.4 Verification

FR-11 After any state change, the system shall re-read authoritative state and produce a field-level before/after diff.

FR-12 The system shall report a clear resolution status: Resolved, Escalated, or Open.

4.5 Failure Handling & Escalation

FR-13 On a transient failure, the system shall retry with backoff and recover automatically.

FR-14 On a permanent failure, the system shall fail fast (no wasted retries) and escalate.

FR-15 The system shall escalate to a human when the order value exceeds a threshold, no automated remedy exists, or a permanent failure occurs. Escalation is a valid terminal outcome.

4.6 Safety & Human Oversight

FR-16 Real outbound notifications shall be sent only when LIVE ACTIONS is enabled; otherwise they are simulated.

FR-17 A financial action at or above the approval threshold shall be flagged approval-required; in strict mode it pauses (awaiting human approval) and mutates nothing until approved.

4.7 Notifications

FR-18 The system shall notify the customer of the outcome on their preferred channel (email or WhatsApp) when enabled.

4.8 Observability

FR-19 The system shall show, per run: the exception identified, tools/systems used, actions performed, result of each action, current resolution status, and whether human intervention is required.

FR-20 The system shall stream each agent step live and provide a per-run metrics summary (tool calls, state changes, retries, LLM turns, duration, provider).

4.9 Operator Platform

FR-21 The operator shall browse all shipments, search, filter by exception, view details, add notes, and run the agent on any shipment.

FR-22 The system shall save each run to history; the operator can re-open, pin, delete, and export a run report.

FR-23 The system shall show insights: resolution rate, escalation rate, average tool calls and duration, and a tool-usage chart.

4.10 Reliability

FR-24 Every tool shall return a structured result and never crash the run.

FR-25 If the primary LLM fails, the system shall fail over to a secondary; if both are unavailable, it shall complete via a deterministic replay of the real tools.

5. Data Entities (simulated system)

Entity	Key fields
Shipment	id, status, carrier, hub, destination, region, ETA, exception_code, attempts
Order	id, customer_id, value, priority, credit
Customer	id, name, email, phone, preferred channel
Carrier	id, regions, capacity, api_status

Entity	Key fields
Inventory	sku, warehouse, quantity
Audit Log	per-action record (tool, ok, attempts, retries, changes)

6. External Interfaces

Interface	Use
Gemini / Groq (LLM)	Tool selection and extraction (failover + rate-paced)
Gemini Vision	OCR of label photos
OpenStreetMap Nominatim	Address validation (with offline fallback)
Resend / Twilio	Email / WhatsApp notifications (gated)
Web UI (Server-Sent Events)	Live streaming of the agent trace

7. Business Rules

- Escalation threshold (order value) and approval threshold (credit amount) are configurable.
- A permanently-failed action is never retried.
- Verification is mandatory before a run is reported complete.

8. Error Handling

- Transient tool failure → retry with backoff; permanent → fail fast → escalate.
- External API down (geocoding / vision) → documented fallback.
- LLM rate-limited or unavailable → failover, then deterministic replay.
- No action mutates state on failure.

9. Assumptions & Dependencies

- Runs against a simulated logistics system (reset per run) plus one real geocoding call.
- Requires an LLM API key (Gemini and/or Groq); notification keys only for LIVE sends.
- Single-worker deployment (in-memory state).

10. Acceptance Criteria

- Three scenarios (email / webhook / photo) each resolve via a different tool path.
- A transient failure recovers; a permanent failure escalates to a terminal state.
- The verified before/after diff reflects a real re-read of state.
- All six visibility items are shown in the UI.
- The smoke test passes (10 / 10) with no real outbound sends.